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Course Content

Assessments

Quiz: Quiz for Week 3 Submitted

Quiz for Week 3

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1. (Special note: Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file [here](#). If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

1 / 1 point

In this problem, we will use the branch-and-bound algorithm to solve the following IP

max 3x1 + 5x2
s.t. x1 + x2 ≤ 16
x2 ≤ 7.5
xi ∈ Z+ i = 1, 2.

As always, our first step is to solve the linear relaxation of the given IP. Write down the unique optimal solution in the order of x1 and x2 by separating two consecutive values with a comma. Do not have any symbol other than numeric values, negation, decimal point, and comma in your answer. For example, if you believe an optimal solution is (0.2, -2.3), write down "0.2,-2.3".

8,5,7,5

Correct

2. Following from the previous problem, branch on x1 to obtain two subproblems. What are the two constraints you add to the first subproblem to generate the two new subproblems?

1 / 1 point

- ☐ x1 = 8 and x1 = 9.
- ☐ x1 ≤ 8 and x1 ≤ 9.
- ☒ x1 ≤ 8 and x1 ≥ 9.
- ☐ x1 ≤ 8 and x2 ≤ 7.
- ☐ None of the above.

Correct

3. Following from the previous problem, find an optimal solution to the original IP. Write down the unique optimal solution in the order of x1 and x2 by separating two consecutive values with a comma. Do not have any symbol other than numeric values, negation, and comma in your answer. For example, if you believe an optimal solution is (2, -3), write down "2,-3".

1 / 1 point

9,7

Correct

4. Recall the following parallel machine makespan minimization problem we introduced in the past. There are m machines and n jobs. For job j, we know its processing time is pj. We want to assign each job to a machine so that the makespan, which is the longest total processing time among all machines, is minimized. For example, suppose that we have n = 15 jobs and m = 3 machines. The processing times are listed in Table 1. These data define an instance for the parallel machine makespan minimization problem.

1 / 1 point

While this problem is NP-hard, it is straightforward to consider the following heuristic algorithm. We simply assign each job one by one to a machine. When a job is to be assigned a machine, it is assigned the machine currently with the smallest total processing time (if there is a tie, choose the machine with the smallest machine ID among those machines having the smallest total process time).

For example, suppose that we assign jobs to machines in the order the job IDs, from small to large. We will then assign job 1 to machine 1, job 2 to machine 2, job 3 to machine 3, and then job 4 to machine 2 (because at that moment machine 2 has the smallest total processing time, which is 4), and then job 5 to machine 3, ..., and finally job 15 to machine 3. The result of applying the simple heuristic algorithm is in Table 2. The makespan of the simple heuristic algorithm on this instance is 45.

Job	Processing time
1	7
2	4
3	6
4	9
5	12
6	6
7	10
8	11
9	8
10	7
11	6
12	8
13	15
14	14
15	3

Table 1: Data for Problem 4

Machine	Jobs assigned to this machine	Total processing time
1	1, 6, 7, 10, 13	7 + 6 + 10 + 7 + 15 = 45
2	2, 4, 8, 11, 14	4 + 9 + 11 + 6 + 14 = 44
3	3, 5, 9, 12, 15	6 + 12 + 8 + 8 + 3 = 37

Table 2: Result of applying the simple heuristic algorithm

Now consider an attempt to improve the above heuristic algorithm. Instead of using job IDs to sort jobs, we instead sort jobs according to their processing times from large to small. When two jobs have the same processing time, we put the one with the smaller job ID in front of the one with the larger job ID. We then apply the same assignment rule (assigning a job to the machine with currently the smallest total processing time) on these jobs one by one according to the predetermined order. In our example, we will first assign job 13 a machine, then job 14, then job 5, ..., and lastly job 15. Please apply the modified heuristic algorithm to this instance and then write down the makespan reported by this heuristic algorithm. Do not have any symbol other than numeric values in your answer.

43

Correct

5. Following from the previous problem, whenever we apply a heuristic algorithm on an instance and obtain a feasible solution, we want to evaluate its performance by calculating the optimality gap. For the instance above, as we do not have an optimal solution, we look for a lower bound on the makespan of an optimal solution. To do this, note that we may formulate an IP for the makespan minimization problem, we may then solve its linear relaxation to obtain a lower bound. For the makespan minimization problem, however, there is an easier way: All we need to do is to divide the total processing time by the number of machines (i.e., to calculate $\frac{\sum_{j \in J} p_j}{m}$, where J is the set of jobs). The ratio is then a lower bound we need. Let z^{ALG} be the makespan you find in the previous problem and z^{LB} be the lower bound you find using the above method, find the (relative) optimality gap $\frac{z^{ALG} - z^{LB}}{z^{LB}}$. Express the optimality gap in percentages and round it down to the closest integer. For example, if you believe that the optimality gap is 0.0678, which is 6.78%, you should write down 6 as your answer.

1 / 1 point

2

Correct